

Question 1: XP Value Identification

A software development team is building an online shopping system. During development, developers continuously discuss requirements with the customer instead of depending only on written documents. Whenever misunderstanding occurs, the team immediately communicates with the customer to clarify the problem.

Identify the Extreme Programming (XP) value demonstrated in this scenario. Explain why this value is important.

Answer

The XP value demonstrated in this scenario is:

Communication

In Extreme Programming (XP), communication emphasizes frequent interaction among developers, customers, and other team members.

Communication is important because:

- It reduces misunderstanding of requirements.
- It allows developers to receive quick clarification.
- It improves collaboration between customers and programmers.
- It prevents project failure caused by missing information.

Therefore, the team's continuous discussion with the customer follows the XP communication principle.

Question 2: XP Process Phase Identification

A development team starts a new project. The customer writes small descriptions of required features on cards. Meanwhile, programmers study the required tools, programming language, and technologies needed for the project.

Which phase of the XP process model is represented here? Describe the activities of this phase.

Answer

The XP process phase represented in this scenario is:

Exploration Phase

In the Exploration Phase:

- Customers create user story cards describing required features.
- Developers understand the technology and tools.
- The team becomes familiar with XP practices.
- Initial understanding of the system requirements is developed.

This phase prepares both customers and developers before actual implementation begins.

Question 3: Pair Programming Scenario

In a banking application project, two programmers work together using the same computer. One programmer writes the code while the other continuously checks mistakes and thinks about improving the solution. After some time, they exchange their responsibilities.

Identify the XP practice used in this scenario and explain the roles.

Answer

The XP practice used in this scenario is:

Pair Programming

Pair programming involves two programmers:

1. Driver

- Controls keyboard and mouse.
- Writes the actual program code.

2. Observer

- Reviews the code continuously.
- Finds errors and suggests improvements.
- Thinks about overall design direction.

Switching roles improves code quality and problem solving.

Question 4: User Story and Planning Scenario

A hospital management software team collects requirements from doctors. The doctors describe required features, developers estimate the required effort, and the most important features are selected for the next release.

Which XP phase is this? Explain the main activities.

Answer

The XP phase described here is:

Planning Phase

Main activities of the Planning Phase:

- User stories are created.
- Developers estimate implementation effort.
- Customers assign priority to requirements.
- Release schedule is prepared.

This phase decides what functionality will be implemented in future iterations.

Question 5: Continuous Integration Scenario

A software company develops a mobile application. Whenever a developer finishes a small part of the program, the new code is immediately combined with the existing system and tested.

Which XP practice is applied here? Explain its benefit.

Answer

The XP practice applied here is:

Continuous Integration

Continuous Integration means adding new completed code into the main system frequently.

Benefits:

- Detects integration problems early.
- Reduces large system conflicts.
- Keeps the software always updated.
- Improves software reliability.

Question 6: XP Death Phase Scenario

A library management system has been successfully completed. The customer has no additional features to request. The system satisfies performance and reliability requirements, and developers prepare final documentation.

Identify the XP lifecycle phase.

Answer

The XP lifecycle phase is:

Death Phase

The Death Phase occurs when:

- No more customer stories remain.
- Customer requirements are completely satisfied.
- Final system documentation is prepared.
- No major changes are expected.

Question 7: XP Role Identification

During an XP project, a team member continuously checks whether the estimated development time matches the actual progress. The person provides feedback about whether the team can complete the iteration within the deadline.

Which XP role is described?

Answer

The XP role described is:

Tracker

Responsibilities of the Tracker:

- Tracks development progress.
- Checks accuracy of effort estimation.
- Provides feedback to improve future estimation.
- Evaluates whether iteration goals are achievable.

The tracker helps the XP team maintain realistic project planning.

Extreme Programming (XP): Tips and Tricks Table

Topic	Memory Trick	Scenario Keywords	Answer Hint
XP Main Idea	FAST + SIMPLE + CUSTOMER	Short cycle, customer involvement, frequent change	Extreme Programming (XP)
XP Values	CSFCR	Team communication, decision, collaboration	XP Values
Communication	Talk More = Less Problems	Discussion, meeting, clarification	Communication
Simplicity	Only Build Needed Features	Avoid unnecessary future requirements	Simplicity
Feedback	Review → Improve → Repeat	Customer review after iteration	Feedback
Courage	Say NO When Needed	Reject unrealistic commitments	Courage
Respect	Care for Team	Teamwork, trust, cooperation	Respect
XP Process Model	E-P-I-P-M-D	Different project stages	XP Lifecycle
Exploration Phase	Explore Before Development	Story cards, learning tools and technology	Exploration
Planning Phase	Plan Before Coding	Priority, effort estimation, release schedule	Planning
Iterations to Release	Build Step by Step	Several iterations, customer selects stories	Iteration Phase
Productionizing	Final Check Before Release	Extra testing, performance checking	Productionizing
Maintenance	Released but Still Improving	Bug fixing, support, new features	Maintenance
Death Phase	Project Completed	No stories left, final documentation	Death
XP Roles	C-P-T-T	Responsibilities mentioned	Identify Role
Customer	Requirement Owner	Writes stories, sets priority	Customer
Programmer	Code Creator	Simple and clear coding	Programmer
Tester	Quality Checker	Functional tests, testing tools	Tester
Tracker	Progress Watcher	Checks estimation and progress	Tracker
Pair Programming	Two Brain One Computer	Two developers working together	Pair Programming
Driver	Hands on Keyboard	Writes actual code	Driver
Observer	Eyes on Code	Reviews code and detects errors	Observer
Continuous Integration	Combine Code Quickly	Immediately adding completed code	Continuous Integration
Small Releases	Release Early, Release Often	Frequent working versions	Small Release
On-site Customer	Customer Always Available	Full-time customer presence	On-site Customer
Coding Standards	Same Coding Style	Common coding rules	Coding Standards
40-Hour Week	Avoid Overload	Limited working hours	40-Hour Week
Open Workspace	Work Together	Shared workspace collaboration	Open Workspace
User Story Card	Customer Wish Card	Short requirement description and priority	User Story Card
Task List	Developer Work Breakdown	Small implementation activities	Task List
Story vs Task	WHAT vs HOW	Requirement vs implementation	Story = Customer, Task = Developer